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On-field kinematics of cut maneuvers in football players: Are wearable sensors reliable for assessing anterior cruciate ligament injury risk?

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ABSTRACT

The aim of the present study was to present the reliability and normative values of wearable inertial sensors kinematics during football-specific cut maneuver tasks collected on the football field for assessing anterior cruciate ligament (ACL) injury risk. Forty-seven academy football players (age 15.9 ± 2.4 years, female $n = 20$) performed a planned 90° change of direction within the Agility T-test and unplanned football-specific changes of direction (FS deceiving action). Kinematics was collected through eight wearable inertial sensors (100 hz, MTw Awinda, Movella). Intraclass correlation coefficient and Root Mean Square Error were used to inspect test-retest and side-to-side reliability of peak and waveform kinematics. Normative kinematics was compared between male and female players (t-test with Cohen's d, $p < 0.05$). Test–retest reliability was moderate-to-excellent in most of the parameters (r=0.40–0.92). Side-to-side reliability was worse than test–retest (both movement tasks). Female players showed worse movement quality than males with greater peak values on the frontal and transverse planes at the knee, pelvis and trunk and smaller knee and trunk flexion ($d = 0.50$–1.1 in Agility T-test, = 0.39–0.73 in FS deceiving action). The on-field cut maneuver kinematics by wearable sensors demonstrated sufficient reliability for most joints. Reliability and normative values might help to objectify ACL injury prevention programs in football academies.

KEYWORDS
Football; change of direction; kinematics; wearable inertial sensors; on-field analysis; injury prevention

Introduction

Musculoskeletal injuries are a huge burden in football medicine. In young football players, the rate of injuries to knee ligaments, in particular to anterior cruciate ligament (ACL), has doubled in the last 10 years (Beck et al., 2017; Weitz et al., 2020). An ACL injury in youth is associated with an 18%–42% rate of second ACL injury (UEFA Champions League) (Della Villa, DiPaolo, et al., 2021), shorter football career and negative impact on physical and mental health. The recent explosion of female football academies could even worsen these outcomes, since a 3–8 times higher ACL injury risk in female compared to male players has been identified USA college (Silvers-Granelli, 2021). Primary prevention actions are therefore necessary to reduce the burden of ACL injuries in the young population.

From a biomechanical point of view, indirect/non-contact injury mechanisms are frequent and mostly happen during defensive situations and pressing (e.g., deceiving actions) (Della Villa et al., 2020; Lucarno et al., 2021). Recurrent injury pattern during high-dynamic, sports-specific movements have been identified as in the studies by Della Villa et al. and Lucarno et al (Della Villa et al., 2020; Lucarno et al., 2021). Indeed, movement biomechanics is becoming crucial in ACL injury prevention programs in football. In particular, data collected in ecological environments has been advocated to improve injury risk patterns detection and bridge the gap between laboratory and sports field (Bolt et al., 2021; Gokeler et al., 2018; Marques et al., 2020).

Maneuvers involving cutting have been identified as the most likely to result in an ACL injury (Della Villa et al., 2020; Dos'Santos et al., 2021; McBurnie et al., 2019). An increased cognitive demand, e.g., the presence of external visual stimuli, sport-specific elements and chaotic environment, has also been demonstrated to affect agility speed performance and biomechanics (Büchel et al., 2022; DiPaolo, Nijmeijer, et al., 2022; Heuvelmans et al., 2024; Rolley et al., 2023). However, the assessment of cutting maneuvers in real-world settings is limited when using traditional motion capture technologies. In response to this limitation, wearable sensors have become increasingly utilized for such analyses (Adesida et al., 2019; Camomilla et al., 2018; DiPaolo, Bragonzoni, et al., 2022; DiPaolo, Nijmeijer, et al., 2023).

Studies have shown that wearable sensors are both reliable and accurate when compared to traditional marker-based motion capture in controlled laboratory settings (Al-Amri et al., 2018; DiPaolo et al., 2021; Nijmeijer et al., 2023; Poitras

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et al., 2019). However, the complexity and variability of player motion in real-world situations present a different set of challenges. Therefore, it is crucial to determine whether the use of wearable inertial sensors, which have been validated primarily in laboratory settings, remains reliable when used to analyze on-field data in football (Gokeler et al., 2022). To the best of the authors' knowledge, no studies have addressed the reliability of biomechanical measurements in football-specific tasks on the field.

The aim of the present study was to provide normative and reliable data of planned and unplanned (football-specific, FS) cut maneuvers collected on the field in young football players. The hypotheses were that (I) wearables sensors would produce at least moderate test-retest and side-to-side reliability in cut maneuvers kinematics, and, if Hp. It could be confirmed that (II) female players would show motion patterns associated to ACL injury mechanism.

Materials and methods

Participants

Forty-seven youth football players (age 15.9 ± 2.4, 27 males, 20 females) were enrolled in the study. During the time the data was collected, each participant in the study was enrolled in a football academy and actively competed in matches at a competitive level. Only outfield players (no goalkeepers) were recruited. Additionally, none of the participants had any musculoskeletal injuries at the time they were included in the study. Detailed demographics are presented in Table 1. Each player's parent/tutor signed informed consent and agreed to their own child's performance data acquisition and treatment for research purposes. This study obtained approval from the *Blinded for submission* (ID: *Blinded for submission*).

Adequate sample size calculation was performed through an a priori power analysis in G*Power (v3.1, Brunsbüttel, Germany). According to the scoping review of Preatoni et al., the median sample size for studies investigating biomechanics from wearables in sports-related musculoskeletal injuries is 17 healthy athletes (multisport) (Preatoni et al., 2022). The only study investigating football-specific agility biomechanics on the field with multiple IMUs collected data on 28 healthy female players (DiPaolo, Nijmeijer, et al., 2022). For the purposes of the present study, a moderate correlation coefficient (r=0.4) was considered with an $\alpha = 0.05$ and power $(1\text{-}\beta) = 0.80$. The minimum required sample size for the present study was therefore estimated in 44 players.

Testing procedure & data collection

The collection of data took place on a football pitch with artificial turf, under daily sunny conditions, during the competitive phase of the season. No testing occurred during the conditioning phase, season break, in rainy weather or on a wet surface in order to minimize potential biases related to player and surface conditions. The players completed both planned 90° changes of direction within the Agility T-test and the unplanned FS changes of direction (FS deceiving action) (Adesida et al., 2019; Marques et al., 2020). A self-paced warm-up (5–10 min) and familiarization trials were conducted before the test (at least two per test, not recorded). The Agility T-test is one of the most common tests in football and multidirectional sports to investigate sprint and agility performance (Krolo et al., 2020; Sugiyama et al., 2021). The test was performance (time)- oriented and did not include any other player/element, as the players were pushed to sprint at their maximum and complete the drill in the shortest time possible. Two trials per side (left side first, non-randomized) were collected for each player. In the FS deceiving action test, participants were engaged in football-specific changes of direction, which involved making a cut towards an opponent holding the ball to imitate a defensive pressing pattern (Figure 1). The players were directed to sprint at their maximum speed possible and try to take the ball away from the opponent as they would in an actual game. They were not informed of which direction the opponent would choose to change. The opponent was standing still 2 m away from the cones and only made the change in direction when the player involved in the test was approaching the cone, in order to simulate a deceptive action (García-Pinillos et al., 2015). Two trials per side were completed in a random order. A rest of 1–2 min in between the trials and 3–5 min in between the two tasks was given to the players.

Lower limbs and trunk kinematics were collected through eight wearable inertial sensors (MTw Awinda, Movella Xsens). The sensors were positioned as follows: foot dorsum (x2), shin (x2), thigh (x2), pelvis (L5), trunk (C7). Two experienced users (*Blinded for submission*) carried on the sensor placement for all the players. Anthropometric measures were collected for each participant before sensors placement, and dynamic calibration of the system (Walk + N Pose) was performed according to the manufacturer's instructions. The output sample frequency was 100 hz.

Table 1. Players' demographics.

N of players	47
Male/Female	27/20
Age (y)	15.9 ± 2.4
Dominant limb (R/L)	45/2
Hrs training/week	4–6
Games/week	1–2

Data processing

Data from four participants were excluded due to technical issues of the wearable inertial sensors system while collecting data. The total number of processed valid trials was 339 (171 for the Agility T-test and 168 for the FS deceiving action). Performance data were extracted for each trial: peak cumulative velocity, peak cumulative acceleration, peak cumulative deceleration, velocity at initial foot contact, acceleration at initial foot contact, high-intensity velocity percentage (HIVP) computed as the percentage of the overall task performed with a speed >4.17 m/s high-intensity acceleration percentage (HIAP) computed as the percentage of the overall task performed with a positive acceleration >2.78 m/s^2, high-intensity deceleration percentage (HIDP) computed as the percentage of the overall task

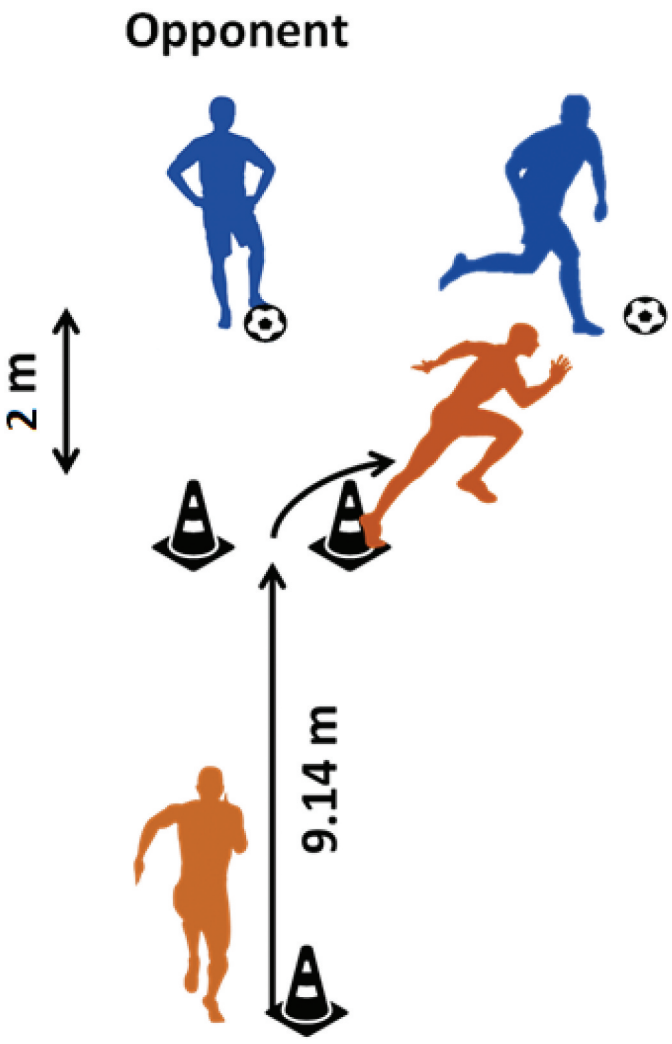

Figure 1. Description of the football-specific deceiving action. The player involved in the test (in orange) sprinted to the cones. The opponent (in blue) suddenly turned left or right (unknown to the player) when the player involved in the test reached the cones. The player involved had to perform an unanticipated cut maneuver to reach the opponent and take the ball away from him/her, as in a common defensive game action.

performed with a negative acceleration $< -2.78\,m/s^2$ (Bastiaansen et al., 2022).

Joint kinematics for ankle, knee, hip, pelvis and trunk in the three anatomical planes were computed for each trial in the company's proprietary software and extracted for post-processing to a custom Matlab script (vR2022a; The MathWorks, Natick, MA, US). The identification of the cut stance phase was performed through the pelvis vertical velocity according to Reenalda et al. (2015) and Milner & Paquette (2015). Data were normalized to include the deceleration and acceleration phase of the cut: the 0% and 100% were, respectively, 25 ms before the beginning and 25 ms after the end of the stance phase of the foot contact in which the players changed their direction (Figure 2) (DiPaolo, Nijmeijer, et al., 2022).

The following features were extracted from each kinematic variable: positive peak angle (e.g., peak hip internal rotation), negative peak angle (e.g., peak hip external rotation), range of motion (ROM), kinematic value at the peak knee flexion angle (to inspect the kinematic response of the body to the load acceptance) (Della Villa, DiPaolo, et al., 2021). Normative kinematics were presented for male and female separately.

Statistical analysis

The normal distribution of the data was inspected through the Shapiro–Wilk test and presented as mean ± standard deviation, while the categorical data were presented as a percentage over the total and median with interquartile range (IQR). The intraclass correlation coefficient (ICC) with the ICC (3,1) model was used to assess the reliability of joint kinematics in test–retest (trial 1 vs trial 2) and side-to-side (dominant vs non-dominant limb). Test-retest and side-to-side reliability were considered excellent for $r > 0.75$, fair for r 0.4–0.74, and poor for $r < 0.39$ (Kadaba et al., 1989). Root mean square error (RMSE) and normalized RMSE (% of the range of motion) were reported alongside the ICC values. The validity was considered excellent for the RMSE (%) < 20% (van der Kruk & Reijne, 2018). Moreover, *T-scores* were computed to establish the pooled reliability of each joint angle under investigation. The analysis of the *T-scores* was necessary to pool validity metrics with different measurement scales and thresholds ($r > 0.4$, RMSE < 20%). This way, it was possible to summarize to what extent a joint angle could be considered reliable during on-field cut maneuver tests with wearable sensors and therefore suitable for injury prevention applications (Bramah et al., 2024; Preatoni et al., 2022; Prill et al., 2021). For each joint angle, RMSE and ICC mean and standard deviation of five kinematics features (positive peak, negative peak, ROM, kinematic value at the peak knee flexion angle and full curve) were extracted from both movement tasks. Then, *T-scores* for RMSE and ICC were calculated according to the following formula: $Tscore = 50 + \left(10 * \frac{Deviation\ Score}{Standard\ Deviation}\right)$, where:

$$- \textit{Deviation score}_{RMSE} = \text{Kinematics feature}_{RMSE} - 20$$

$$- \textit{Deviation score}_{ICC} = \text{Kinematics feature}_{ICC} - 0.4$$

The *T-score* formula was converted from z-score formula. The first element of the formula (constant: 50) is chosen as the mean value point on a scale 0–100 and the second element is *z-score* *10, where *Deviation score* is the deviation from the validity metrics ($r = 0.4$ and RMSE = 20%, respectively). *T-scores* = 50, *T-score* ≤40 and *T-score* ≥60 are equivalent to threshold, 1 standard deviation below the threshold and 1 standard deviation above the threshold, respectively. *T-scores* were interpreted as ≤45 good-to-excellent, >45 to ≤55 moderate, >55 poor (Bramah et al., 2024).

The two-tailed Student's t-test was used to compare continuous kinematics and performance variables between males and females, and Cohen's d effect size was reported. Cohen's d values were reported as a measure of the effect size, where $0.2 \le d \le 0.5$, $0.5 \le d \le 0.8$ and $d \ge 0.8$ represent a small, moderate and large effect, respectively (Cohen, 2013). Differences were reported with 95% with confidence interval (CI). The Mann-Whitney U test was used to investigate the differences in non-normally distributed variables. Differences were considered statistically significant for $p < 0.05$. For the sake of conciseness, the discussion will focus on the variables

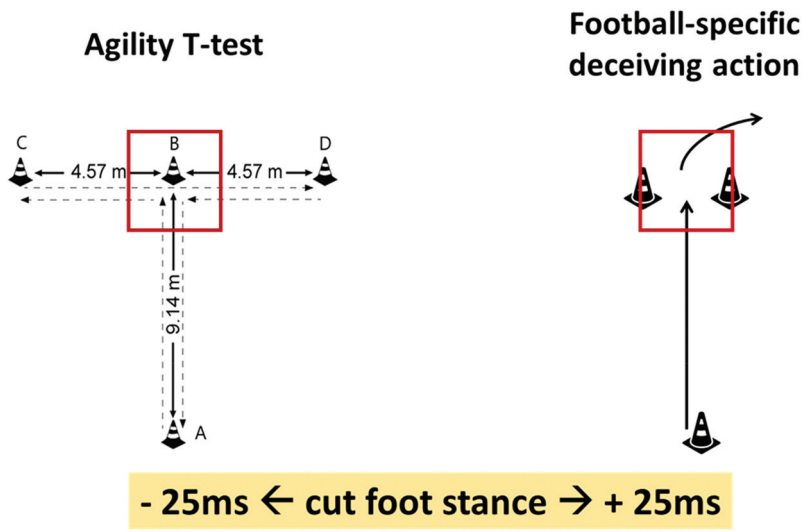

Figure 2. Description of the area under investigation for the kinematics assessment of the football players. The assessment window includes 25 ms before to 25 ms after the stance phase of the foot contact of the cut maneuver. In the Agility T-test (planned cut maneuver, left) the assessment is performed on the first 90° change of direction. In the football-specific deceiving action (unplanned cut maneuver, right), the assessment is performed at the time of the move from the opponent.

obtaining at least moderate effect size (d > 0.5), moderate reliability ($r > 0.4$) and excellent RMSE (<20%). The statistical analyses were performed in Matlab.

Results

Reliability analysis

In Agility T-test, test–retest reliability was excellent at ankle, pelvis and trunk peak flexion (r=0.80–0.92) and at pelvis and hip transverse plane at peak and peak knee flexion angle, respectively (r=0.76–0.77). Moderate test–retest reliability was found at peak and peak knee flexion in most of the other parameters, except knee internal-external rotation (r=0.40–0.74). Full curve test–retest reliability was moderate (r=0.41–0.72). Side-to-side reliability was excellent at pelvis and trunk peak and ROM flexion (r=0.80–0.88), but overall poor-to-moderate in the other parameters (r=0.0–0.69) and at the curve level (r=0.05–0.71, Table 2, Appendix A1.1). Median RMSE was 6.2° (IQR 5.6°) and 19% (IQR 6%, Figure 3).

In FS deceiving action, test–retest reliability was excellent at pelvis and trunk flexion, hip extension, peak knee varus and knee valgus and rotation at peak knee flexion angle (r=0.75–0.98). Moderate test–retest reliability was found in most of the other parameters (r=0.42–0.74). Side-to-side reliability was excellent at pelvis and trunk peak and ROM flexion (r=0.76–0.81), mostly poor in the peak and peak knee flexion parameters (r=0.0–0.68) and mostly moderate at ROM and curve level (r=0.05–0.66, Table 2, Appendix A1.2). Median RMSE was 6.3° (IQR 5.3°) and 19% (IQR 3%, Figure 3).

Pooled reliability of each joint angle (pooled ICC and RMSE *T-scores* for kinematics of both movements) is described in Figure 4 and Appendix A2.

Normative kinematics

During the Agility T-test (planned cut maneuver), female players showed a greater peak knee external rotation (d=1.1, $p < 0.001$), greater knee, pelvis and trunk ROM on the frontal plane

Table 2. Test–retest (TRT) and side-to-side (STS) reliability in Agility T-test and football-specific deceiving action. Complete kinematics features results (positive peak, negative peak, range of motion) were presented in appendix A1.1-A1.2. *Note*: RMSE = root mean square error, ICC = intraclass correlation coefficient.

		Agility T-test						Football-specific deceiving action					
		Kinematics at peak knee flexion angle			Full curve			Kinematics at peak knee flexion angle			Full curve		
Joint	Angle (°)	RMSE (%)	ICC_{STS}	ICC_{TRT}	RMSE (%)	ICC_{STS}	ICC_{TRT}	RMSE (%)	ICC_{STS}	ICC_{TRT}	RMSE (%)	ICC_{STS}	ICC_{TRT}
Ankle	Abd-add	6.9 (23)	0.0	0.40	8.2 (18)	0.58	0.53	7.5 (21)	0.31	0.54	9.0 (18)	0.52	0.46
	Int-ext	10.7 (16)	0.24	0.63	10.4 (18)	0.47	0.55	9.9 (21)	0.34	0.48	11.1 (20)	0.41	0.50
	Flex-ext	10.7 (16)	0.0	0.17	11.1 (20)	0.30	0.42	10.8 (22)	0.27	0.70	11.6 (22)	0.40	0.51
Knee	Abd-add	5.0 (14)	0.35	0.40	3.4 (18)	0.31	0.44	4.2 (16)	0.55	0.97	3.7 (18)	0.37	0.44
	Int-ext	3.3 (14)	0.0	0.12	3.4 (17)	0.33	0.41	3.2 (17)	0.36	0.75	3.4 (18)	0.36	0.42
	Flex-ext	13.1 (23)	0.32	0.61	16.5 (22)	0.29	0.42	18.2 (23)	0.43	0.58	14.2 (20)	0.36	0.47
Hip	Abd-add	7.0 (16)	0.0	0.48	6.8 (18)	0.04	0.41	6.0 (20)	0.0	0.42	7.1 (20)	0.43	0.46
	Int-ext	9.4 (16)	0.33	0.77	9.3 (18)	0.34	0.51	8.4 (19)	0.22	0.59	8.5 (18)	0.28	0.60
	Flex-ext	12.5 (23)	0.40	0.70	12.4 (20)	0.45	0.57	12.9 (21)	0.42	0.34	14.6 (22)	0.29	0.56
Pelvis	Abd-add	3.3 (26)	0.06	0.48	3.3 (21)	0.71	0.63	3.6 (26)	0.11	0.61	3.9 (25)	0.62	0.56
	Int-ext	2.4 (19)	0.54	0.63	2.9 (18)	0.46	0.43	2.9 (20)	0.04	0.59	4.0 (20)	0.44	0.57
	Flex-ext	4.4 (24)	0.77	0.87	4.8 (22)	0.60	0.72	4.4 (23)	0.78	0.86	4.7 (19)	0.66	0.50
Trunk	Abd-add	5.3 (21)	0.0	0.42	5.2 (20)	0.36	0.36	6.0 (20)	0.0	0.33	6.4 (20)	0.54	0.46
	Int-ext	5.4 (19)	0.0	0.28	6.4 (17)	0.05	0.43	6.4 (20)	0.0	0.33	7.4 (19)	0.17	0.45
	Flex-ext	9.8 (24)	0.78	0.87	10.7 (22)	0.60	0.72	9.8 (23)	0.78	0.86	10.6 (19)	0.66	0.50

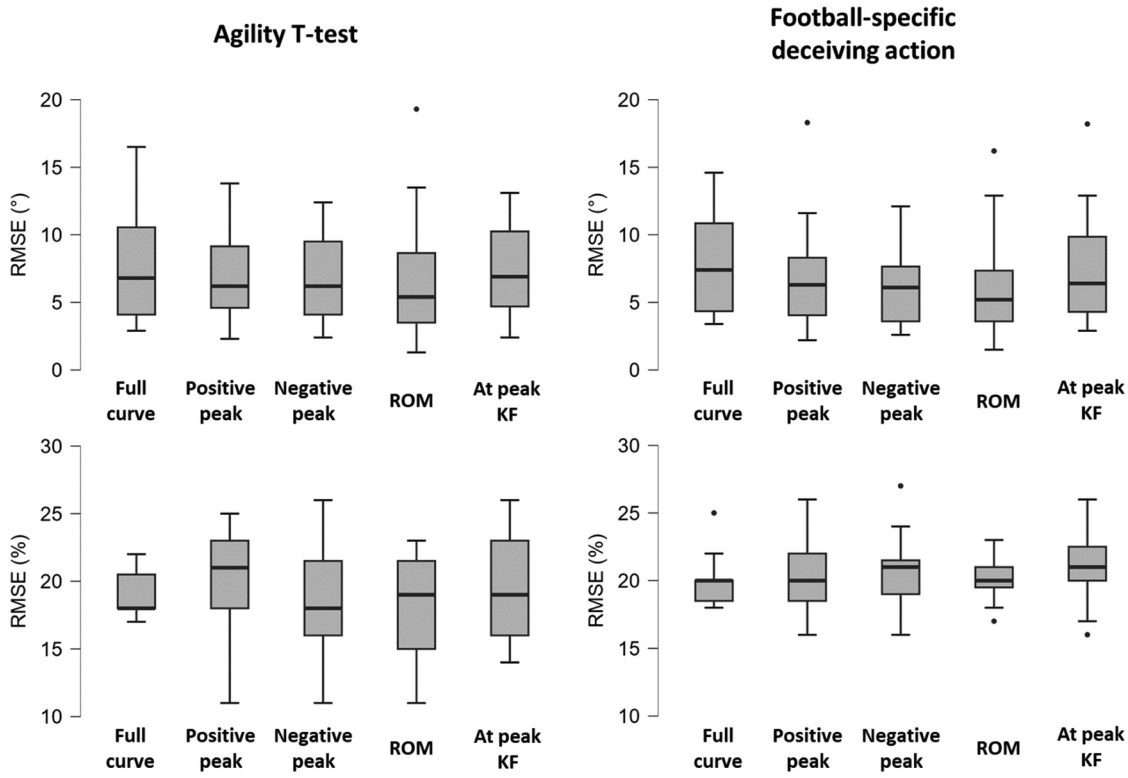

Figure 3. Root mean square error (RMSE, top row) and normalized RMSE (bottom row) for Agility T-test and football-specific deceiving action for all the features analyzed. Median RMSE was between 5–10° and around 20% for all the features.

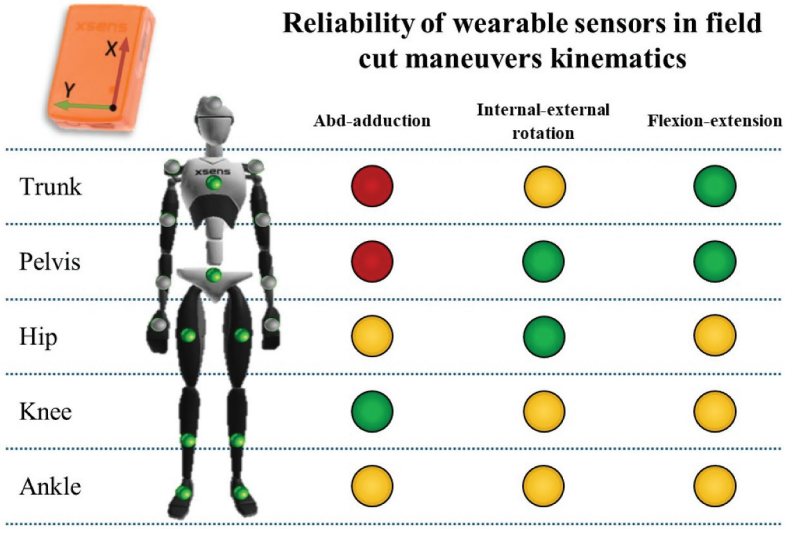

Figure 4. Pooled reliability of joint kinematics parameters collected through wearable inertial sensors during cut maneuvers on the field. Green, yellow and red dots represent joint angles with good-to-excellent, moderate and poor reliability, respectively, according to intraclass correlation coefficient and normalized root mean square error *T-scores*. Detailed description of single *T-scores* is presented in appendix A2.

(d=0.57–0.59, $p < 0.001$), lower knee, pelvis and trunk flexion at the peak knee flexion angle (d=0.50–0.58, $p < 0.002$), peak hip external rotation and knee valgus (d=0.22–0.45, $p = 0.006$–0.138) compared to their male counterparts (Table 3, Appendix A3.1).

In the FS deceiving action (unplanned cut maneuver), female players showed greater peak knee internal rotation, lower trunk tilt and extension (d=0.54–0.73, $p < 0.001$) and lower knee flexion ROM and at the peak knee flexion angle (d=0.39–0.42, $p = 0.011$–0.018) than male players (Table 3, Appendix A3.2).

Table 3. Normative kinematics (angles in degrees) of the Agility T-test and football-specific (FS) deceiving action. Complete kinematics features results (positive peak, negative peak, range of motion) were presented in appendix A2.1-A2.2. *Note*: positive rotations: abduction, internal rotation, flexion.

Joint	Angle (°)	All	Male	Female	p-value	d	Diff 95% CI
Agility T-test (kinematics at peak knee flexion angle)							
Ankle	Abd-add	−5.7 ± 8.4	−5.5 ± 8.1	−6.1 ± 8.9	0.566	0.09	[−1.9;3.4]
	Int-ext	−6.2 ± 12.6	−6.3 ± 10.4	−6.1 ± 15.4	0.288	0.16	[−13.1;3.9]
	Flex-ext	−7.3 ± 13.2	−7.5 ± 11.7	−7.0 ± 15.4	0.824	0.03	[−4.7;3.8]
Knee	Abd-add	−1.1 ± 6.3	−1.5 ± 4.7	−0.4 ± 8.1	0.866	0.03	[−1.5;1.3]
	Int-ext	−0.3 ± 4.0	0.1 ± 3.0	−1.0 ± 5.1	0.008	0.40	[−7.6;-1.2]
	Flex-ext	77.5 ± 14.9	80.9 ± 13.0	72.4 ± 16.2	<0.001	0.58	[3.9;13.1]
Hip	Abd-add	7.9 ± 7.9	8.3 ± 6.6	7.2 ± 9.6	0.05	0.31	[0;4.4.0]
	Int-ext	5.7 ± 10.3	6.8 ± 8.8	3.9 ± 11.9	0.111	0.24	[−4.0;0.4]
	Flex-ext	35.2 ± 14.2	36.9 ± 13.7	32.6 ± 14.7	0.065	0.30	[−0.3;8.7]
Pelvis	Abd-add	0.6 ± 3.6	0.4 ± 3.8	1.0 ± 3.3	0.667	0.07	[−1.3;0.9]
	Int-ext	−0.3 ± 2.8	−0.1 ± 2.7	−0.6 ± 2.9	0.537	0.10	[−0.5;0.9]
	Flex-ext	6.9 ± 4.7	7.9 ± 4.1	5.5 ± 5.2	0.002	0.50	[0.9;3.8]
Trunk	Abd-add	5.2 ± 6.3	6.3 ± 5.6	3.5 ± 6.8	0.005	0.46	[0.8;4.3]
	Int-ext	−1.0 ± 6.2	−1.0 ± 6.0	−1.0 ± 6.5	0.526	0.10	[−1.0;2.0]
	Flex-ext	15.5 ± 10.4	17.6 ± 9.1	12.4 ± 11.6	0.002	0.50	[2.0;8.5]
FS deceiving action (kinematics at peak knee flexion angle)							
Ankle	Abd-add	−6.3 ± 9.1	−6.9 ± 8.2	−5.3 ± 10.4	0.157	0.23	[−4.7;0.8]
	Int-ext	−4.2 ± 11.2	−4.1 ± 11.1	−4.3 ± 11.6	0.548	0.10	[−1.9;3.6]
	Flex-ext	−14.4 ± 12.6	−16 ± 12.3	−11.9 ± 12.9	0.044	0.33	[−8.2;-0.1]
Knee	Abd-add	−2.7 ± 4.5	−3.5 ± 4.3	−1.5 ± 4.6	0.592	0.09	[−0.8;1.4]
	Int-ext	−1.3 ± 3.5	−0.7 ± 3.6	−2.2 ± 3.2	<0.001	0.68	[−4.3;-1.6]
	Flex-ext	93.1 ± 19.6	96.1 ± 20.4	88.8 ± 17.6	0.021	0.39	[1.1;13.6]
Hip	Abd-add	7.5 ± 7.0	8.3 ± 6.2	6.5 ± 7.9	0.570	0.09	[−1.7;3.1]
	Int-ext	1.8 ± 9.1	3.6 ± 9.5	−0.7 ± 8.1	0.632	0.08	[−1.7;2.8]
	Flex-ext	20.0 ± 15.9	20.6 ± 15.4	19.3 ± 16.6	0.614	0.08	[−3.8;6.4]
Pelvis	Abd-add	0.2 ± 3.9	0.3 ± 4.2	0.1 ± 3.4	0.597	0.09	[−0.9;1.6]
	Int-ext	−0.5 ± 3.3	0.0 ± 3.4	−1.3 ± 3.1	0.427	0.13	[−1.7;0.7]
	Flex-ext	5.4 ± 4.7	5.4 ± 4.4	5.4 ± 5.1	0.965	0.01	[−1.5;1.5]
Trunk	Abd-add	5.6 ± 6.6	7.0 ± 6.2	3.4 ± 6.5	<0.001	0.73	[2.3;6.1]
	Int-ext	0.3 ± 7.5	0.9 ± 7.4	−0.6 ± 7.6	0.486	0.11	[−3.7;1.7]
	Flex-ext	12.0 ± 10.5	11.9 ± 9.8	12.1 ± 11.5	0.943	0.01	[−3.5;3.3]

Average waveform kinematics is presented in Figure 5, extensive normative kinematics is presented in Appendix A4.1-A4.2.

Performance analysis

Female players showed lower velocity at both peak and initial foot contact (d = 0.64–1.09, $p < 0.001$) in both tasks compared to male players (Table 4). Peak acceleration and deceleration were also higher in male players compared to female players with a small effect in Agility T-test (d = 0.10–0.32, $p = 0.044$–-0.514) and moderate effect in FS deceiving action (d = 0.46–0.55, $p = 0.001$–0.008). HIAP and HIDP ranged between 23.5% and 34.0% in male players and 5.2–15.2% in female players (d = 0.64–1.53, $p < 0.001$).

Discussion

In the current investigation, our objective was to establish reliability and normative kinematic measurements for two cut maneuver tasks performed on the football field by young football players. We utilized wearable inertial sensors to analyze the kinematics for male and female participants, focusing on planned (Agility T-test) and unplanned (FS deceiving action) cut maneuvers with specific football-related elements. The study consisted of test-retest and side-to-side reliability assessments for multiple kinematic features, encompassing five features (positive peak, negative peak, range of motion, kinematics at peak knee flexion angle, full curve) for each of the five joints (ankle, knee, hip, pelvis, trunk) on the three anatomical planes. This data was collected to gain insight into the most reliable measurements for assessing players' cutting motion in a real-world, ecological setting.

The most important findings of the present study were that (I) test–retest reliability was acceptable (moderate-to-excellent) for most of the joints and planes in ACL injury-related features, and that (II) female athletes exhibited (in both movement tasks) biomechanical risk factors in line with the ACL injury mechanisms from the literature (Lucarno et al., 2021).

Test–retest reliability was acceptable for most of the joints (Figure 4). Overall, sagittal plane reliability was the best in both test-retest and side-to-side. Knee and hip flexion were also moderately valid in both movement tasks, as well as hip internal-external rotation and knee abd-adduction. However, although frontal and transverse plane reliability were also mostly valid in terms of test-retest, worse outcomes emerged when assessing side-to-side differences. Despite the fact that none of the players had a history of musculoskeletal injury at lower limbs, cut maneuvers performed with dominant and non-dominant side (on the field) might show differences that should be taken into account when pooling the data and when comparing injured players with 'healthy' controls or contralateral limbs. This was clearer when assessing sagittal plane kinematics, probably due to the generally higher range of motion of these features. Inter-limb differences might definitely be sport-specific and be accentuated in football, where dominant and non-dominant sides are used differently during the game, especially at the non-professional level (Brophy et al., 2010; DeLang et al., 2021).

Median RMSE was below 6.5° and 20% of the range of motion in both movement tasks. According to previous studies

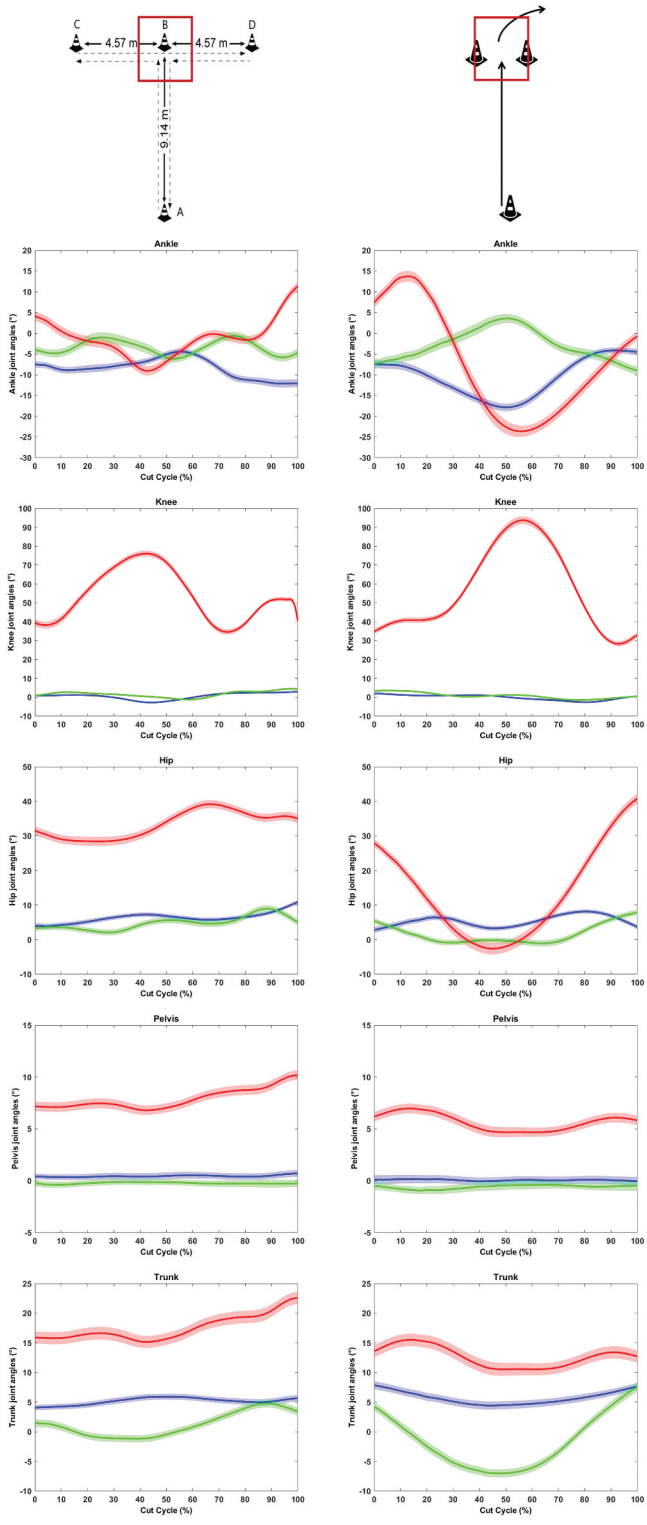

Figure 5. Average kinematics of the first change of direction of the Agility T-test (planned cut maneuver, left) and of the football-specific deceiving action (unplanned cut maneuver, right). Data are presented as mean (solid line) and standard deviation (shaded area). Sagittal plane kinematics is presented in red, frontal plane kinematics in blue, transverse plane kinematics in green. *Note*: positive rotations: abduction, internal rotation, flexion.

inspecting the concurrent validation of wearable sensors technology against the gold standard and in increasing complexity scenarios, this aspect should be interpreted as a proof of validity of the measure in most of the joint parameters under investigation (DiPaolo et al., 2021; Nijmeijer et al., 2023; Poitras et al., 2019; Prill et al., 2021). However, caution should be used when interpreting small differences between players (e.g., between healthy and ACL injured players) as both

Table 4. Players' performance in the two tasks. *Note*: FC = foot contact; HIVP: high-intensity velocity percentage; HIDP: high-intensity deceleration percentage; HIAP: high-intensity acceleration percentage.

Parameter	All	Male	Female	p-value	d	Diff [95% CI]
Agility *T*- test						
Time to complete (s)	14.7 ± 2.8	12.9 ± 1.7	17.1 ± 1.9	<0.001	2.32	[3.1;5.3]
Peak velocity (m/s)	5.5 ± 0.7	5.8 ± 0.4	5.0 ± 0.9	<0.001	1.09	[0.5;1.0]
Peak acceleration (m/s^2)	8.0 ± 6.2	8.8 ± 4.6	6.7 ± 7.9	0.044	0.31	[0.1;4.0]
Peak deceleration (m/s^2)	−8.2 ± 5.4	−8.4 ± 4.7	−7.8 ± 6.4	0.514	0.10	[−2.3;1.2]
Velocity at FC (m/s)	3.9 ± 0.9	4.2 ± 0.8	3.6 ± 1.0	<0.001	0.64	[0.3;0.9]
Acc. at FC (m/s^2)	2.4 ± 5.3	3.1 ± 5.8	1.5 ± 4.3	0.058	0.32	[−0.1;3.3]
HIVP (%)	66.2 ± 22.7	67.0 ± 16.3	64.9 ± 29.8	0.565	0.09	[−5.1;9.4]
HIAP (%)	24.4 ± 20.0	34.0 ± 18.9	10.3 ± 11.1	<0.001	1.53	[18.5;28.9]
HIDP (%)	20.4 ± 14.7	24 ± 15.1	15.2 ± 12.3	<0.001	0.64	[4.3;13.3]
FS Deceiving action						
Time to complete (s)	3.9 ± 0.7	3.6 ± 0.6	4.3 ± 0.7	<0.001	1.07	[0.3;1.1]
Peak velocity(m/s)	5.6 ± 0.7	5.9 ± 0.6	5.1 ± 0.5	<0.001	1.43	[0.6;1.0]
Peak acceleration (m/s^2)	8.1 ± 5.4	9.2 ± 6.0	6.4 ± 3.8	0.001	0.55	[1.1;4.4]
Peak deceleration (m/s^2)	−7.4 ± 5.9	−8.5 ± 6.8	−5.8 ± 3.8	0.007	0.47	[−4.5;-0.7]
Velocity at FC (m/s)	5.2 ± 0.9	5.5 ± 0.9	4.7 ± 0.7	<0.001	0.96	[0.5;1.1]
Acc. at FC (m/s^2)	3.3 ± 5.5	4.3 ± 6.3	1.9 ± 3.8	0.008	0.46	[0.6;4.1]
HIVP (%)	73.9 ± 17.0	73.5 ± 14.3	74.6 ± 20.5	0.698	0.06	[−6.6;4.4]
HIAP (%)	21.5 ± 20.7	28.2 ± 22.6	11.6 ± 12.0	<0.001	0.92	[10.4;22.7]
HIDP (%)	16.1 ± 20.3	23.5 ± 23.2	5.2 ± 5.2	<0.001	1.09	[12.5;24.2]

significantly and clinically relevant, especially when regarding transverse plane kinematics (e.g., at the knee and ankle level). This aspect is in line with the current literature expressing more confidence in sagittal plane kinematics at all joints and decreasing confidence on the frontal/transverse plane variables with task complexity increase (Al-Amri et al., 2018; DiPaolo et al., 2021; Nijmeijer et al., 2023).

We combined the reliability and RMSE of the two movement tasks across all features to determine which variables were sufficiently reliable (good-to-excellent reliability according to *T-scores*), moderately reliable (moderate reliability according to *T-scores*), or not sufficiently reliable (poor reliability according to *T-scores*) for evaluating on-field movements using wearable inertial sensors (Figure 4). All sagittal plane parameters were either sufficiently or moderately reliable as well as ankle, knee and hip frontal and transverse plane parameters, while frontal plane pelvis and trunk should be considered not sufficiently reliable. It should be noted that this classification is based on the thresholds of excellence as reported in the *Statistical Analysis* section (Bramah et al., 2024; Kadaba et al., 1989; van der Kruk & Reijne, 2018).

Differences were observed between male and female players in all planes. Specifically, males performed better (speed, acceleration, elapsed time) in the two tasks and exhibited safer movement quality during the cut according to the ACL injury mechanisms literature (Della Villa et al., 2020; Sigurðsson et al., 2020). For frontal and transverse plane, females displayed greater peak knee abduction and external rotation than male counterparts, as well as peak hip external rotation, suggesting a greater tendency to dynamic knee valgus (DiPaolo, Grassi, et al., 2023; Lucarno et al., 2021; Sigurðsson et al., 2020). These differences were consistent across planned and unplanned cuts, particularly in terms of knee rotation and trunk tilt. For the sagittal plane, female players exhibited lower peak knee flexion and lower flexion of the upper body than male players. These findings were also common between the two tasks. The adoption of a stiff sagittal strategy has been commonly recognized as a risk factor for ACL injury due to the limited shock absorption and the higher intra-articular forces that can be generated (Leppänen et al., 2017; Pollard et al., 2010). Both prevention and rehabilitation programs indeed focus on the adoption of greater knee, hip and trunk flexion during cut maneuvers to favor the shock absorption at muscle level (Alentorn-Geli et al., 2009; Dos'Santos, Thomas, et al., 2019; Markström et al., 2018; Nagelli et al., 2019).

The reduced number of significant differences found in the unplanned cut compared to the pre-planned cut might mean an increase of inter-subject variability (larger standard deviation) and the adoption of risky biomechanical patterns also by males. The presence of sport-specific elements contributes to distract the player from the movement execution and favors the implementation of the usual movement schemes that are adopted during trainings and matches as according to the ecological dynamics approach (Bolt et al., 2021; Büchel et al., 2022). However, this could also favor the presence of biomechanical patterns that put the knee joint at higher risk of overload, as also previously reported (DiCesare et al., 2020; Nijmeijer et al., 2023; Peel et al., 2019). Testing players' movement in game-resembling conditions seems therefore a crucial upgrade of the common testing procedures in order to shift the player's focus from the movement execution to a performance- and sport-related goal (e.g., regain the ball from your opponent) and to amplify the chances to see unfavorable patterns – and eventually work with the player to correct them.

The reliability measures and the differences emerged between males and females suggest that the analysis of single kinematics values in the assessment of ACL injury risk is not recommended. On the other hand, an analysis including the multiplanar complexity of the movement patterns and of the intra- and inter-joint interaction during the cut maneuver might be more beneficial in the overall comprehension of the quality of a player's movement with specific regard to the type of injury pattern under investigation. The adoption of multiplanar scores and frameworks has been already proposed and validated in different contexts (lab, field) by means of video-analysis technique and wearable inertial sensors (Della Villa, DiPaolo, et al., 2021; DiPaolo, Bragonzoni, et al., 2022; Dos'Santos, McBurnie, et al., 2019; Lloyd, 2021) and has been

used in the assessment of ACL injury risk and return to sport after surgery.

The adoption of complex sport-specific movement tasks in ecological environment can challenge the players both in terms of performance and injury risk and should be considered when assessing players' movement through quantitative measurements. The present study was the first to assess both planned and unplanned cut maneuvers in a football environment by means of wearable inertial sensors. No spatial constraints were applied to the players during their motion, and game-resembling situations were inspected. Normative data were provided as well as the errors and reliability measures of the single joint measures. The present study also puts emphasis on the limits of the wearable sensor technology in the assessment of high-dynamics movement task with specific regards to ACL injury prevention. The present study also showed one of the greatest sample sizes of the current literature: most of the studies had indeed a sample size of 10 participants for females and 15 participants for males, according to the systematic review by Preatoni and colleagues (Preatoni et al., 2022). Normative and reliability values as presented in the present paper might be of help to future researchers involved in the processing and interpretation of complex field movement data and could have a final use in implementing objective testing for ACL injury prevention through quantitative measures (Büchel et al., 2022; Gokeler et al., 2022; Lloyd, 2021; Preatoni et al., 2022).

The present study has some limitations. First, the design was cross-sectional, thus it was not possible to assess the day-by-day reliability of the wearable inertial sensors and to inspect differences in movement pattern among the players. Furthermore, only one sensor type, validated against marker-based motion capture in high-dynamics movements, was adopted (Al-Amri et al., 2018; DiPaolo et al., 2021; Nijmeijer et al., 2023; Poitras et al., 2019). The adoption of multiple sensor types could have proved the superiority of one system on the others in these applications. Second, only healthy young players were included in the study: future analysis should investigate the differences in biomechanics between healthy and ACL-injured players cleared for return to sport. Kinematics differences among players according to their roles in the field were not investigated as well as the relation between functional outcomes (strength tests, VO2max, etc.) and kinematics. It should be also acknowledged that normative and reliable results might not be generalizable to adult players. Factors such as level of play (professional vs non-professional), years of football practice and playing style could influence the findings, which might be extended to adult non-professional players the most. Third, due to the sport-specific and unplanned nature of the FS deceiving action, this movement task could not be standardized in terms of speed and cut angle. Agility T-test and unplanned cut maneuvers are common to multiple sports (basketball, rugby, American football, etc.). Thus, the present study's findings could be extended to other sports. However, caution should be adopted when generalizing to other types of cutting maneuvers (e.g., other agility tasks). Last, the present study mostly focused on peak kinematics and investigated waveforms only in part. Future analyses should focus on overall movement patterns either within the same joint and anatomical plane and between multiple joint angles (joint coordination).

Conclusion

Wearable inertial sensors showed good reliability in test-retest and side-to-side comparisons on joint kinematics in planned and unplanned cutting maneuvers performed in the football field. However, it is important to interpret joint angle values cautiously, and a comprehensive approach that considers multiple parameters, such as injury profiles are advisable. Establishing reliability and normative values can assist researchers and practitioners in developing and refining injury prevention programs for ACL injuries in football academies.

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Author contributions

All the authors actively contribute to the development of the present work. In particular:

- Di Paolo S. contributed to the study design, developed and conducted the motion capture data collection, post-processed the data, structured and interpreted the results and wrote a substantial part of the manuscript
- Gokeler A. contributed to the study design, interpreted the results and contributed to manuscript writing and editing.
- Benjaminse contributed to study design, interpreted the results and contributed to manuscript writing and editing.
- Zaffagnini S. contributed to study design, results interpretation and presentation and supervised the entire manuscript editing process.
- Bragonzoni L. contributed to study design, contributed to data collection and supervised results interpretation, presentation and manuscript editing process.

Institutional review board approval

Bioethical Committee of the University of Bologna (IRB approval: n. 25861 of 10 February 2020)

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